LEARNING-BASED DATA FUSION

Pedestrian detection with 3 types of data fusion

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Objective: compare three fusion methods

no learning-based data fusion

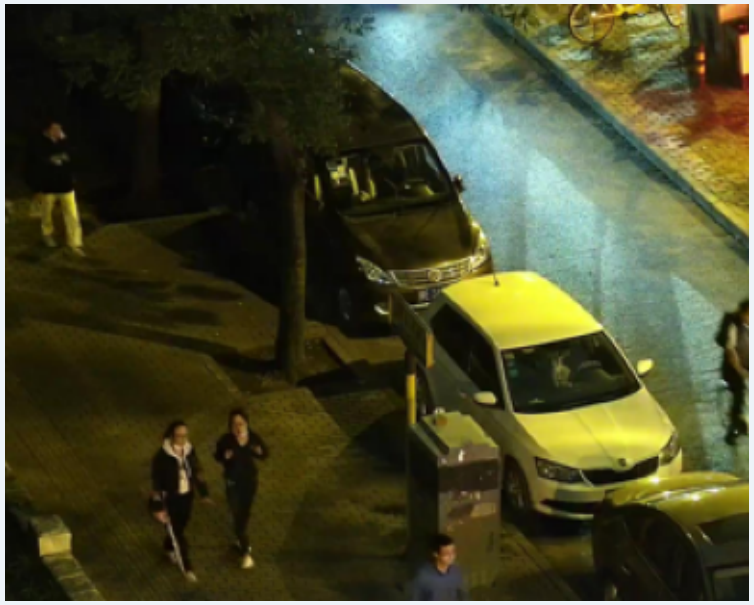
learning-based data fusion

fine tuned learning-based data fusion

Do the network and fine-tuning help **detect more pedestrians**

compared with non-learning-based data fusion?

Dataset: AWMM-100K and two image types



Visible: colour camera



Thermal: infrared camera

AWMM-100K: pretraining. Images shown: LLVIP, used for our fine-tuning.

Artificial rain in our experiment

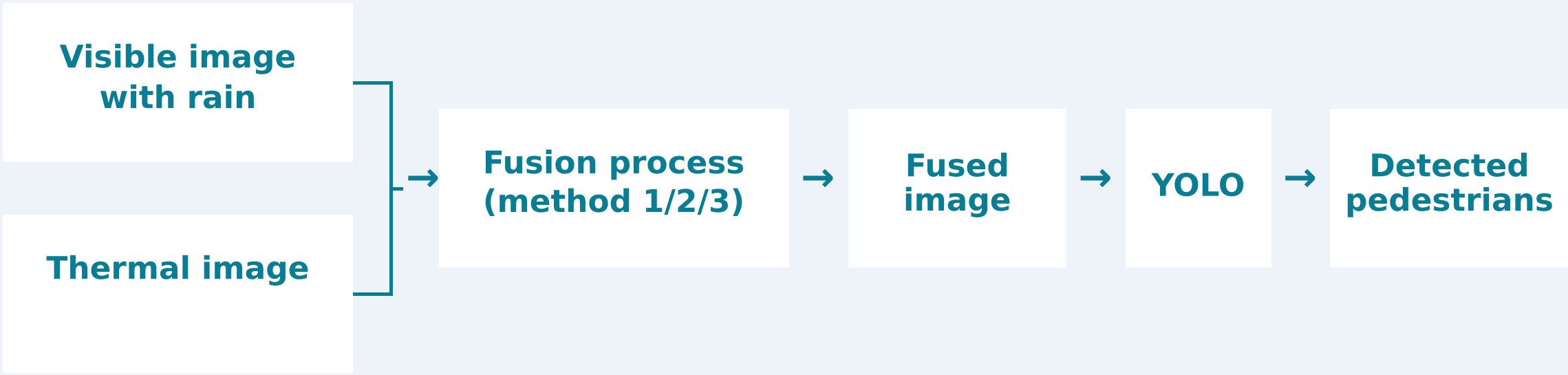
In our experiment, artificial rain is added only to the colour camera images.



Goal: compare the 3 types of data fusion using

the image with rain and the infrared camera image

The complete pipeline:



1) No learning-based data fusion

Rule: average the corresponding pixels of the two images

$$\text{Fused image} = \frac{\text{Visible} + \text{Thermal}}{2}$$

Each pixel receives half its contribution from each image.
The weights are the same everywhere: the fusion does not learn.

2) Learning-based data fusion

- Pretrained network used: **AMG-Fuse**
- Dataset used: AWMM-100K with 1,000 image pairs
- AMG-Fuse processes visible and thermal inputs to directly produce an image **R**

1. Extracts features	→	2. Combines them	→	3. Reconstructs
Finds edges and contrasts in the two inputs.		Combines information through attention.		Produces an initial combined image **R**

How the network guides fusion

The code takes R, then computes α by comparing it with the two inputs.

- R closer to visible → $\alpha > 0.5$ → greater visible contribution.
- R closer to thermal → $\alpha < 0.5$ → greater thermal contribution.

$$\text{Fused image} = \alpha \times \text{Visible} + (1 - \alpha) \times \text{Thermal}$$

Fixed average: α = 0.5 everywhere. Here, the weight varies with the scene and position.

Fine-tuned learning-based data fusion

- We start with the same AMG-Fuse and adapt only the **last layer**
- **LLVIP dataset** : 100 image pairs
- 3 epochs: **2,592** updated weights

1. Detection	→	2. Error	→	3. Update
Fused image → YOLO → pedestrian boxes.		Comparison with the ground truth boxes in the dataset.		The error updates only the final fusion layer.

YOLO: from fused images to pedestrians

Each fused image from the three methods is fed separately into the same YOLO.

Fused image → YOLO → pedestrian boxes + scores

Threshold = minimum score required to show a detection.

Higher threshold ↑

Fewer boxes shown.
May reduce false alarms,
but also miss real pedestrians.

Lower threshold ↓

More boxes shown.
May recover pedestrians,
but also add false alarms.

Interface

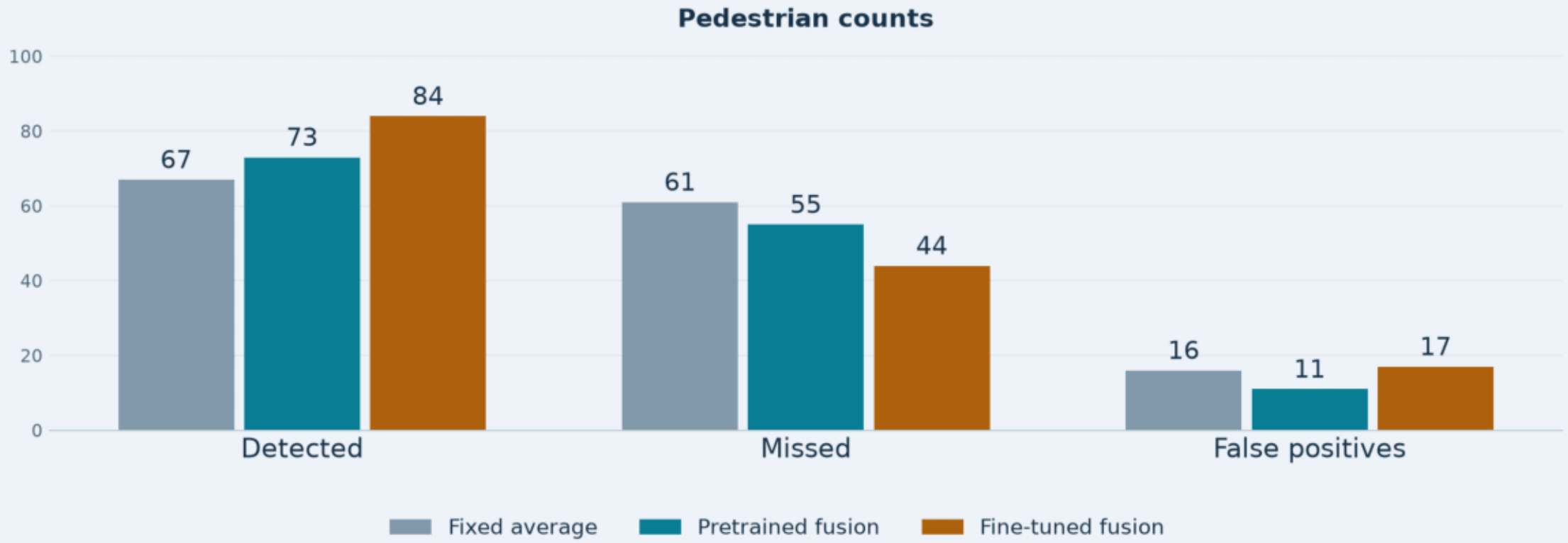
We now compare the three methods on the same **50** image pairs

from the LLVIP test set using the **SAME YOLO**

http://127.0.0.1:64001

Performance of the three fusion methods

50 LLVIP test pairs · 128 annotated pedestrians · same frozen YOLO



Two data fusion problems and their solutions

1. Giving equal weight to images of different quality

I used spatially varying weights guided by the network.

2. Risking the loss of pedestrian details

I applied the enlarged weight map to the original images.

Three things I learned from this project

1. Combining two sensors

2. Adapting a pretrained network

3. Evaluating an improvement

Thank you for your attention